

3.5.4 Homeostasis is the maintenance of a constant internal environment.

Principles	Homeostasis in mammals involves physiological control systems that maintain the internal environment within restricted limits. The importance of maintaining a constant core temperature and constant blood pH in relation to enzyme activity. The importance of maintaining a constant blood glucose concentration in terms of energy transfer and water potential of blood.
Temperature control	The contrasting mechanisms of temperature control in an ectothermic reptile and an endothermic mammal. Mechanisms involved in heat production, conservation and loss. The role of the hypothalamus and the autonomic nervous system in maintaining a constant body temperature in a mammal.
Control of blood glucose concentration	The factors that influence blood glucose concentration. The role of the liver in glycogenesis and gluconeogenesis. The role of insulin and glucagon in controlling the uptake of glucose by cells and in activating enzymes involved in the interconversion of glucose and glycogen. The effect of adrenaline on glycogen breakdown and synthesis. The second messenger model of adrenaline and glucagon action. Types I and II diabetes and control by insulin and manipulation of the diet.